CHAPTER 6

STATICS AND STIFFNESS ANALYSIS

6.1 INTRODUCTION

When a manipulator performs a given task, such as lifting up a workpiece from an NC machine as shown in Fig. 6.1, its end effector exerts force and moment to the external environment at the point(s) of contact. This force and moment are generated by actuators installed at various points of connection. For brevity, we often use the term *force* to imply both force and moment. For a serial manipulator, actuator forces are transmitted through an open-loop chain to the point of contact, while for a parallel manipulator, actuator forces are transmitted through several parallel paths to the end effector.

Static force analysis is of practical importance in determining the quality of force transmission through the various joints of a mechanism. It serves as a basis for sizing the links and bearings of a robot manipulator and for selecting appropriate actuators. The results can also be used for compliance control of a robot manipulator. The statics of spatial mechanisms can be treated by various methods, such as the vector method (Beyer, 1960; Chace, 1962), the dual vector and dual number quaternions (Yang, 1965), the screw calculus (Agrawal and Roth, 1992; Bagci, 1971; Denavit et al., 1965), and the principle of virtual work (Asada and Slotine, 1986; Gosselin, 1990; Paul, 1981; Tahmasebi and Tsai, 1995).

In this chapter we develop methods for representing static forces acting on a manipulator and for transforming them between coordinate systems. The transformation of force and moment between the actuator space and the end-effector space is the focal point of study. We show that the actuator input

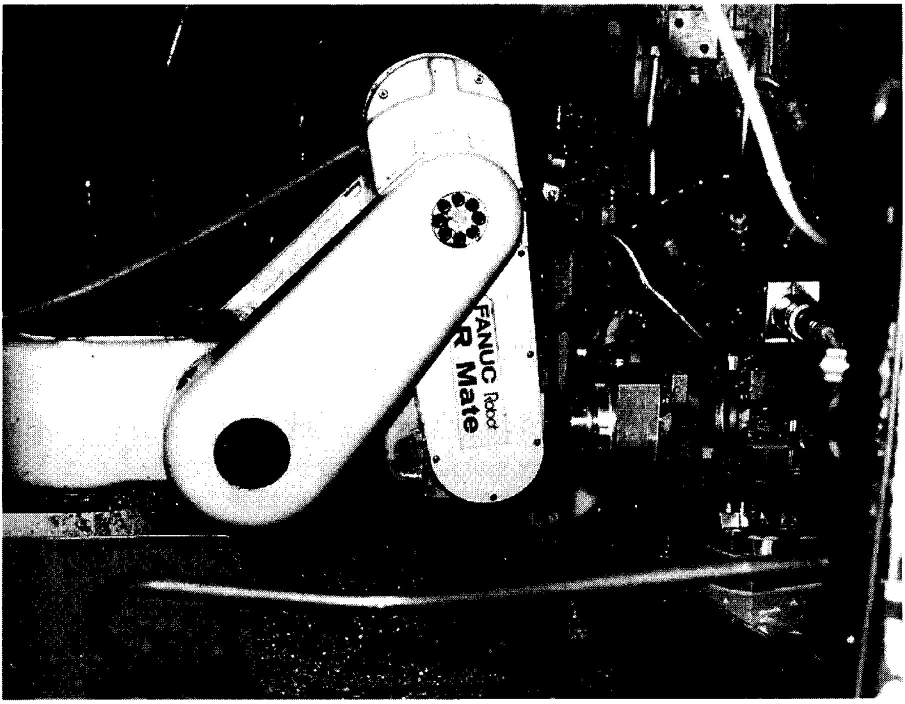

FIGURE 6.1. Fanuc LR Mate robot. (Courtesy of Fanuc Robotics North America, Inc., Rochester Hills, Michigan.)

forces are related to the end-effector output forces by the transpose of the manipulator Jacobian matrix. In addition, the free-body diagram method is introduced for the derivation of reaction forces generated at the various joints of a manipulator. A thorough understanding of the joint reaction forces is important for proper sizing of links and actuators at the design stage. Static force analysis is followed by stiffness analysis. It is shown that the end-effector output forces are related to its deflections by a matrix called the *stiffness matrix*. Both serial and parallel manipulators are studied.

6.2 STATICS OF SERIAL MANIPULATORS

In this section we study the statics of serial manipulators. First, we derive the basic equations governing the static balance of a link. Then these equations are applied for the static analysis of serial manipulators. The concept of equivalent joint torques and the transformation between the end-effector forces and equivalent joint torques are described.

6.2.1 Force and Moment Balance of a Link

In a serial manipulator, each link is connected to one or two other links by various joints. Figure 6.2 depicts the forces and moments acting on a typical link i that is connected to link $i-1$ by joint i and to link $i+1$ by joint $i+1$. The forces acting on link $i+1$ by link i in joint $(i+1)$ can be reduced to a resultant force $\mathbf{f}_{i+1,i}$ and a resultant moment $\mathbf{n}_{i+1,i}$ about the origin O_i of the (x_i, y_i, z_i) link coordinate frame. Similarly, the forces acting on link i by link $i-1$ in the ith joint can be reduced to a resultant force $\mathbf{f}_{i,i-1}$ and a moment $\mathbf{n}_{i,i-1}$ about the origin O_{i-1} of the $(x_{i-1}, y_{i-1}, z_{i-1})$ link coordinate frame. The following notations are defined:

$\mathbf{f}_{i+1,i}$: resulting force exerted on link $i+1$ by link i at O_i, $\mathbf{f}_{i,i+1} = -\mathbf{f}_{i+1,i}$.

$\mathbf{g}$: acceleration of gravity.

m_i: mass of link i.

$\mathbf{n}_{i+1,i}$: resulting moment exerted on link $i+1$ by link i, about point O_i, $\mathbf{n}_{i,i+1} = -\mathbf{n}_{i+1,i}$.

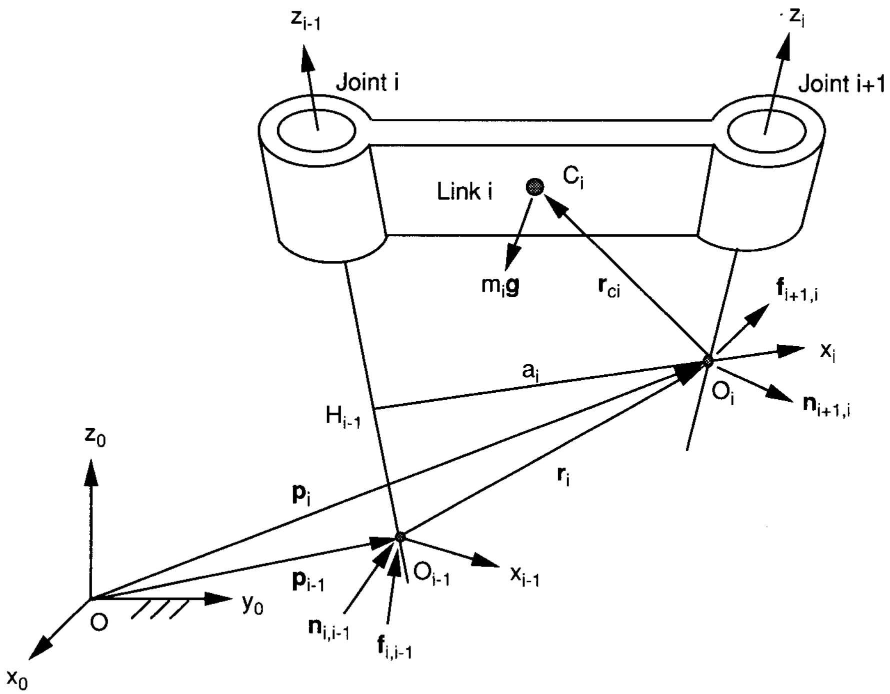

FIGURE 6.2. Forces and moments acting on link i.

$\mathbf{r}_{ci}$: position vector of the center of mass of link i relative to the ith link frame (i.e., $\mathbf{r}_{ci} = \overline{O_i C_i}$).

$\mathbf{r}_i$: position vector of O_i with respect to the $(i-1)$th link frame (i.e., $\mathbf{r}_i = \overline{O_{i-1} O_i}$).

First we consider the balance of forces. As shown in Fig. 6.2, there are three forces exerted on link i: $\mathbf{f}_{i,i-1}$, $-\mathbf{f}_{i+1,i}$, and $m_i\mathbf{g}$. Hence a force balance equation can be written as

$$\mathbf{f}_{i,i-1} - \mathbf{f}_{i+1,i} + m_i \mathbf{g} = \mathbf{0}. \tag{6.1}$$

Next, we consider the balance of moments about the origin O_i. As shown in Fig. 6.2, there are two moments acting on link i: $\mathbf{n}_{i,i-1}$ and $-\mathbf{n}_{i+1,i}$. In addition, the two forces $m_i\mathbf{g}$ and $\mathbf{f}_{i,i-1}$ produce moments about O_i. Summing these moments together, we obtain

$$\mathbf{n}_{i,i-1} - \mathbf{n}_{i+1,i} - \mathbf{r}_i \times \mathbf{f}_{i,i-1} + \mathbf{r}_{ci} \times m_i \mathbf{g} = \mathbf{0}. \tag{6.2}$$

We call $\mathbf{f}_{i+1,i}$ and $\mathbf{n}_{i+1,i}$ the reaction force and moment between link i and link $i+1$. For $i = 0$, $\mathbf{f}_{1,0}$ and $\mathbf{n}_{1,0}$ represent the force and moment exerted on the first moving link by the base link. For $i = n$, $\mathbf{f}_{n+1,n}$ and $\mathbf{n}_{n+1,n}$ represent the force and moment exerted on the environment by the end effector. In this regard, the environment is treated as an additional link, numbered $n + 1$.

Equations (6.1) and (6.2) written once for every moving link, $i = 1, 2, \ldots, n$, yield $2n$ vector equations in $2(n + 1)$ number of reaction forces and moments. Therefore, to yield a unique solution, two of the reaction forces and moments should be specified. When a manipulator performs a given task, such as insertion or grinding, the end effector exerts some force and/or moment to its environment. On the other hand, when a manipulator carries an object, the weight of the object becomes a load to the end effector. Hence by considering the end-effector output force and moment, $\mathbf{f}_{n+1,n}$ and $\mathbf{n}_{n+1,n}$, as known, Eqs. (6.1) and (6.2) can be solved for the remaining reaction forces and moments.

For convenience, we often combine the force and moment into a six-dimensional vector,

$$\mathbf{F}_{i,i-1} = \begin{bmatrix} \mathbf{f}_{i,i-1} \\ \mathbf{n}_{i,i-1} \end{bmatrix}. \tag{6.3}$$

We call the foregoing six-dimensional vector $\mathbf{F}_{i,i-1}$ a *wrench* about O_{i-1}.

6.2.2 Recursive Method

In this section we develop a recursive method for the static force analysis of serial manipulators. The recursive method solves joint reaction forces and moments one link at a time without the need of solving $2n$ vector equations simultaneously. To facilitate the analysis, we write Eqs. (6.1) and (6.2) in the following recursive form:

$$\mathbf{f}_{i,i-1} = \mathbf{f}_{i+1,i} - m_i\mathbf{g}, \tag{6.4}$$

$$\mathbf{n}_{i,i-1} = \mathbf{n}_{i+1,i} + \mathbf{r}_i \times \mathbf{f}_{i,i-1} - \mathbf{r}_{ci} \times m_i\mathbf{g}. \tag{6.5}$$

The vectors in Eqs. (6.4) and (6.5) are expressed in the fixed frame. However, the position vector $\mathbf{r}_{ci}$ is often specified in the ith link frame. Similarly, the vector $\mathbf{r}_i$ can conveniently be expressed, in terms of the D-H parameters, in the ith link frame as

$$^i\mathbf{r}_i = \begin{bmatrix} a_i \\ d_i s\alpha_i \\ d_i c\alpha_i \end{bmatrix}. \tag{6.6}$$

Therefore, both $^i\mathbf{r}_{ci}$ and $^i\mathbf{r}_i$ should be transformed into the fixed frame before they are substituted into Eqs. (6.4) and (6.5). This can easily be done by the following transformations:

$$\mathbf{r}_{ci} = {}^0R_i\, {}^i\mathbf{r}_{ci}, \tag{6.7}$$

$$\mathbf{r}_i = {}^0R_i\, {}^i\mathbf{r}_i. \tag{6.8}$$

Applying Eqs. (6.4) and (6.5), the joint reaction forces can be computed recursively. The process starts at the end-effector link, one link at a time, and ends at the base link. For $i = n$, the end-effector output force and moment, $\mathbf{f}_{n+1,n}$ and $\mathbf{n}_{n+1,n}$, are considered as known. Hence Eqs. (6.4) and (6.5) give the reaction force and moment, $\mathbf{f}_{n,n-1}$ and $\mathbf{n}_{n,n-1}$, at the nth joint. The process is repeated for $i = n-1, n-2, \ldots, 1$ until all the reaction forces are found.

Alternatively, we can compute the joint reaction forces in each individual link frame. Writing Eqs. (6.4) and (6.5) in the ith link frame, we obtain

$$^i\mathbf{f}_{i,i-1} = {}^i\mathbf{f}_{i+1,i} - m_i\, {}^i\mathbf{g}, \tag{6.9}$$

$$^i\mathbf{n}_{i,i-1} = {}^i\mathbf{n}_{i+1,i} + {}^i\mathbf{r}_i \times {}^i\mathbf{f}_{i,i-1} - {}^i\mathbf{r}_{ci} \times m_i\, {}^i\mathbf{g}. \tag{6.10}$$

Once the reaction forces are computed in the ith link frame, they are converted into the $(i-1)$th link frame by the following transformations:

$$^{i-1}\mathbf{f}_{i,i-1} = {}^{i-1}R_i \, {}^{i}\mathbf{f}_{i,i-1}, \tag{6.11}$$

$$^{i-1}\mathbf{n}_{i,i-1} = {}^{i-1}R_i \, {}^{i}\mathbf{n}_{i,i-1}. \tag{6.12}$$

Note that the ${}^{i}\mathbf{g}$ term in Eqs. (6.9) and (6.10) denotes the acceleration of gravity expressed in the ith link frame. Since $\mathbf{g}$ is usually specified in the fixed frame, it should be converted into the link frame before it is substituted into Eqs. (6.9) and (6.10). This can be accomplished by the following recursive formula. For $i = 0$, ${}^{0}\mathbf{g}$ is known. For $i = 1$ to n, we compute

$$^{i}\mathbf{g} = {}^{i}R_{i-1} \, {}^{i-1}\mathbf{g} \tag{6.13}$$

in sequence. Furthermore, if the end-effector output force and moment are specified in the fixed frame, they should also be transformed into the end effector frame:

$$^{n}\mathbf{f}_{n+1,n} = {}^{n}R_0 \, {}^{0}\mathbf{f}_{n+1,n}, \tag{6.14}$$

$$^{n}\mathbf{n}_{n+1,n} = {}^{n}R_0 \, {}^{0}\mathbf{n}_{n+1,n}. \tag{6.15}$$

Equations (6.4) and (6.5) compute the joint reaction forces in the fixed frame, while Eqs. (6.9) and (6.10) calculate the same forces in the link frames. Both are equally useful for the static analysis of serial manipulators.

6.2.3 Equivalent Joint Torques

Once the reaction forces in the joints are known, the actuator forces and/or torques can be determined. For a serial manipulator, each joint is driven by an actuator that exerts either a force or a torque between two adjacent links. These actuator forces and/or torques can be found by projecting the reaction forces onto their corresponding joint axes.

For a prismatic joint, the actuator force is exerted along the ith joint axis. Assuming that frictional force at the joint is negligible, the actuator force, τ_i, is given by

$$\tau_i = \mathbf{z}_{i-1}^{\mathrm{T}} \mathbf{f}_{i,i-1}, \tag{6.16}$$

where $\mathbf{z}_{i-1}$ is a unit vector pointing along the positive ith joint axis. Equation (6.16) implies that the actuator only bears the component of $\mathbf{f}_{i,i-1}$ along the direction of the joint axis, while the other components of $\mathbf{f}_{i,i-1}$ are supported by the joint bearings.

Similarly, for a revolute joint, the actuator exerts a torque instead of force about the ith joint axis. This actuator torque, τ_i, is given by

$$\tau_i = \mathbf{z}_{i-1}^{\mathrm{T}} \mathbf{n}_{i,i-1}. \tag{6.17}$$

Again, the actuator only carries the component of $\mathbf{n}_{i,i-1}$ along the direction of the joint axis, while other components of $\mathbf{n}_{i,i-1}$ are supported by the bearings. We call τ_i the equivalent joint torque.

Example 6.2.1 *Statics of a Planar 3-DOF Manipulator* Figure 6.3 shows the planar $3R$ manipulator studied in Chapters 2 and 4. A coordinate system with all the z-axes pointing out of the paper is defined for each link according to the D-H convention. Let the end-effector output force and moment be given by $\mathbf{f}_{4,3} = [f_x, f_y, 0]^T$ and $\mathbf{n}_{4,3} = [0, 0, n_z]^T$, respectively. Also let the acceleration of gravity, $\mathbf{g}$, be pointing along the negative y_0-direction and the center of mass be located at the midpoint of each link. We wish to find the joint reaction forces and moments.

The D-H parameters and transformation matrices are given in Table 2.1 and Eqs. (2.6) through (2.8). The vectors ${}^i\mathbf{r}_i$ and ${}^i\mathbf{r}_{ci}$ are

$$ {}^i\mathbf{r}_i = \begin{bmatrix} a_i \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad {}^i\mathbf{r}_{ci} = \begin{bmatrix} -a_i/2 \\ 0 \\ 0 \end{bmatrix}. \tag{6.18} $$

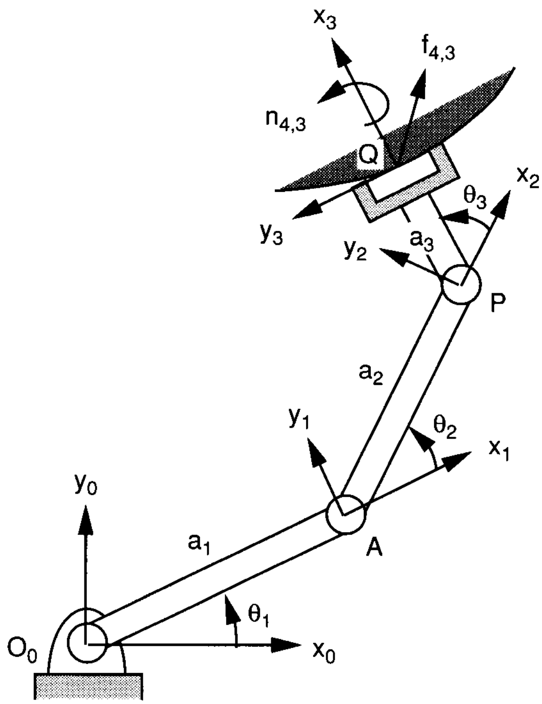

FIGURE 6.3. Planar $3R$ manipulator exerting a force $f_{4,3}$ and a moment $n_{4,3}$.

Substituting Eq. (6.18) for $i = 1$ to 3 into Eqs. (6.7) and (6.8) gives

$$\mathbf{r}_1 = {}^0R_1\,{}^1\mathbf{r}_1 = a_1 \begin{bmatrix} c\theta_1 \\ s\theta_1 \\ 0 \end{bmatrix}, \quad \mathbf{r}_{c1} = {}^0R_1\,{}^1\mathbf{r}_{c1} = -\frac{a_1}{2} \begin{bmatrix} c\theta_1 \\ s\theta_1 \\ 0 \end{bmatrix}.$$

$$\mathbf{r}_2 = {}^0R_2\,{}^2\mathbf{r}_2 = a_2 \begin{bmatrix} c\theta_{12} \\ s\theta_{12} \\ 0 \end{bmatrix}, \quad \mathbf{r}_{c2} = {}^0R_2\,{}^2\mathbf{r}_{c2} = -\frac{a_2}{2} \begin{bmatrix} c\theta_{12} \\ s\theta_{12} \\ 0 \end{bmatrix}.$$

$$\mathbf{r}_3 = {}^0R_3\,{}^3\mathbf{r}_3 = a_3 \begin{bmatrix} c\theta_{123} \\ s\theta_{123} \\ 0 \end{bmatrix}, \quad \mathbf{r}_{c3} = {}^0R_3\,{}^3\mathbf{r}_{c3} = -\frac{a_3}{2} \begin{bmatrix} c\theta_{123} \\ s\theta_{123} \\ 0 \end{bmatrix}.$$

We now apply Eqs. (6.4) and (6.5) to compute the reaction forces exerted on link 3, then proceed to link 2 and 1 in sequence. For $i = 3$, substituting $\mathbf{r}_3$, $\mathbf{r}_{c3}$, $\mathbf{f}_{4,3}$, and $\mathbf{n}_{4,3}$ into Eqs. (6.4) and (6.5) yields

$$\mathbf{f}_{3,2} = \mathbf{f}_{4,3} - m_3\mathbf{g} = \begin{bmatrix} f_x \\ f_y + m_3 g_c \\ 0 \end{bmatrix},$$

$$\mathbf{n}_{3,2} = \mathbf{n}_{4,3} + \mathbf{r}_3 \times \mathbf{f}_{3,2} - \mathbf{r}_{c3} \times m_3\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ n_{3,2z} \end{bmatrix},$$

where

$$n_{3,2z} = n_z + f_y a_3 c\theta_{123} - f_x a_3 s\theta_{123} + 0.5 m_3 g_c a_3 c\theta_{123}.$$

For $i = 2$, we substitute $\mathbf{f}_{3,2}$ and $\mathbf{n}_{3,2}$ obtained in the preceding step along with $\mathbf{r}_2$ and $\mathbf{r}_{c2}$ into Eqs. (6.4) and (6.5). As a result, we obtain

$$\mathbf{f}_{2,1} = \mathbf{f}_{3,2} - m_2\mathbf{g} = \begin{bmatrix} f_x \\ f_y + (m_2 + m_3) g_c \\ 0 \end{bmatrix},$$

$$\mathbf{n}_{2,1} = \mathbf{n}_{3,2} + \mathbf{r}_2 \times \mathbf{f}_{2,1} - \mathbf{r}_{c2} \times m_2\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ n_{2,1z} \end{bmatrix},$$

where

$$\begin{aligned} n_{2,1z} &= n_z + f_y(a_2 c\theta_{12} + a_3 c\theta_{123}) - f_x(a_2 s\theta_{12} + a_3 s\theta_{123}) \\ &\quad + 0.5 m_2 g_c a_2 c\theta_{12} + m_3 g_c(a_2 c\theta_{12} + 0.5 a_3 c\theta_{123}). \end{aligned}$$

For $i = 1$, we substitute $\mathbf{f}_{2,1}$ and $\mathbf{n}_{2,1}$ obtained in the preceding step along with $\mathbf{r}_1$ and $\mathbf{r}_{c1}$ into Eqs. (6.4) and (6.5). This produces

$$\mathbf{f}_{1,0} = \mathbf{f}_{2,1} - m_1\mathbf{g} = \begin{bmatrix} f_x \\ f_y + (m_1 + m_2 + m_3)g_c \\ 0 \end{bmatrix},$$

$$\mathbf{n}_{1,0} = \mathbf{n}_{2,1} + \mathbf{r}_1 \times \mathbf{f}_{1,0} - \mathbf{r}_{c1} \times m_1\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ n_{1,0z} \end{bmatrix},$$

where

$$\begin{aligned} n_{1,0z} = n_z &+ f_y(a_1 c\theta_1 + a_2 c\theta_{12} + a_3 c\theta_{123}) - f_x(a_1 s\theta_1 + a_2 s\theta_{12} + a_3 s\theta_{123}) \\ &+ 0.5 m_1 g_c a_1 c\theta_1 + m_2 g_c(a_1 c\theta_1 + 0.5 a_2 c\theta_{12}) \\ &+ m_3 g_c(a_1 c\theta_1 + a_2 c\theta_{12} + 0.5 a_3 c\theta_{123}). \end{aligned}$$

Finally, we apply Eq. (6.17) to compute the joint torques as follows:

$$\begin{aligned} \tau_1 &= \mathbf{z}_0^{\mathrm{T}}\, \mathbf{n}_{1,0} = n_{1,0z}, \\ \tau_2 &= \mathbf{z}_1^{\mathrm{T}}\, \mathbf{n}_{2,1} = n_{2,1z}, \\ \tau_3 &= \mathbf{z}_2^{\mathrm{T}}\, \mathbf{n}_{3,2} = n_{3,2z}. \end{aligned}$$

We note that in the absence of gravity, the torques and end-effector output forces are related by the following equation:

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = J^{\mathrm{T}} \begin{bmatrix} f_x \\ f_y \\ n_z \end{bmatrix}, \tag{6.19}$$

where

$$J = \begin{bmatrix} -(a_1 s\theta_1 + a_2 s\theta_{12} + a_3 s\theta_{123}) & -(a_2 s\theta_{12} + a_3 s\theta_{123}) & -a_3 s\theta_{123} \\ (a_1 c\theta_1 + a_2 c\theta_{12} + a_3 c\theta_{123}) & (a_2 c\theta_{12} + a_3 c\theta_{123}) & a_3 c\theta_{123} \\ 1 & 1 & 1 \end{bmatrix}.$$

Hence, in the absence of gravity, the transformation between the end-effector output forces and the joint torques is governed by the transpose of the conventional Jacobian matrix.

Example 6.2.2 *Statics of the Stanford Manipulator* As a second example, we consider the Stanford arm shown in Fig. 4.8. Let the end-effector output force and moment be denoted as $\mathbf{f}_{7,6} = [f_x, f_y, f_z]^{\mathrm{T}}$ and $\mathbf{n}_{7,6} = [n_x, n_y, n_z]^{\mathrm{T}}$, respectively. Also let the acceleration of gravity be pointing in the negative z_0-direction (i.e., $\mathbf{g} = [0, 0, -g_c]^{\mathrm{T}}$). To simplify the analysis, we assume that the masses of links 1, 2, 4, 5, and 6 are negligible, and the center of mass of link 3 is located at the midpoint of the link. We wish to find the joint reaction forces.

The D-H parameters and the transformation matrices are given in Chapter 4. The unit vectors, $\mathbf{z}_{i-1}$ for $i = 0$ to 5 are given in Eqs. (4.69) through (4.73). Some relevant ${}^i\mathbf{r}_i$ and ${}^i\mathbf{r}_{ci}$ vectors are as follows:

$$
{}^2\mathbf{r}_2 = \begin{bmatrix} 0 \\ d_2 \\ 0 \end{bmatrix}, \qquad {}^3\mathbf{r}_3 = \begin{bmatrix} 0 \\ 0 \\ d_3 \end{bmatrix}, \qquad {}^3\mathbf{r}_{c3} = \begin{bmatrix} 0 \\ 0 \\ -d_3/2 \end{bmatrix},
$$

$$
{}^1\mathbf{r}_1 = {}^4\mathbf{r}_4 = {}^5\mathbf{r}_5 = {}^6\mathbf{r}_6 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.
$$

Transforming the vectors above into the fixed frame by Eqs. (6.7) and (6.8), we obtain

$$
\mathbf{r}_2 = {}^0R_2\,{}^2\mathbf{r}_2 = d_2 \begin{bmatrix} -\mathrm{s}\theta_1 \\ \mathrm{c}\theta_1 \\ 0 \end{bmatrix},
$$

$$
\mathbf{r}_3 = {}^0R_3\,{}^3\mathbf{r}_3 = d_3 \begin{bmatrix} \mathrm{c}\theta_1 \mathrm{s}\theta_2 \\ \mathrm{s}\theta_1 \mathrm{s}\theta_2 \\ \mathrm{c}\theta_2 \end{bmatrix},
$$

$$
\mathbf{r}_{c3} = {}^0R_3\,{}^3\mathbf{r}_{c3} = -\frac{d_3}{2} \begin{bmatrix} \mathrm{c}\theta_1 \mathrm{s}\theta_2 \\ \mathrm{s}\theta_1 \mathrm{s}\theta_2 \\ \mathrm{c}\theta_2 \end{bmatrix},
$$

$$
\mathbf{r}_1 = \mathbf{r}_4 = \mathbf{r}_5 = \mathbf{r}_6 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.
$$

We now apply Eqs. (6.4) and (6.5) to compute the reaction forces one link at a time beginning from link 6. Due to negligible masses of links 4, 5, and 6, and the concurrence of the fourth, fifth, and sixth joint axes, the reaction forces at the last three joints are equal to one another; that is,

$$\mathbf{f}_{6,5} = \mathbf{f}_{5,4} = \mathbf{f}_{4,3} = \mathbf{f}_{7,6} = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix},$$

$$\mathbf{n}_{6,5} = \mathbf{n}_{5,4} = \mathbf{n}_{4,3} = \mathbf{n}_{7,6} = \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}.$$

For $i = 3$, substituting $\mathbf{r}_3$, $\mathbf{r}_{c3}$, $\mathbf{f}_{4,3}$, and $\mathbf{n}_{4,3}$ into Eqs. (6.4) and (6.5) yields

$$\mathbf{f}_{3,2} = \mathbf{f}_{4,3} - m_3\mathbf{g} = \begin{bmatrix} f_x \\ f_y \\ f_z + m_3 g_c \end{bmatrix},$$

$$\begin{aligned} \mathbf{n}_{3,2} &= \mathbf{n}_{4,3} + \mathbf{r}_3 \times \mathbf{f}_{3,2} - \mathbf{r}_{c3} \times m_3\mathbf{g} \\ &= \begin{bmatrix} n_x + f_z d_3 s\theta_1 s\theta_2 - f_y d_3 c\theta_2 + 0.5 m_3 g_c d_3 s\theta_1 s\theta_2 \\ n_y - f_z d_3 c\theta_1 s\theta_2 + f_x d_3 c\theta_2 - 0.5 m_3 g_c d_3 c\theta_1 s\theta_2 \\ n_z + f_y d_3 c\theta_1 s\theta_2 - f_x d_3 s\theta_1 s\theta_2 \end{bmatrix}. \end{aligned}$$

For $i = 2$, we substitute $\mathbf{f}_{3,2}$ and $\mathbf{n}_{3,2}$ obtained in the preceding step along with $\mathbf{r}_2$ and $\mathbf{r}_{c2}$ into Eqs. (6.4) and (6.5). As a result, we obtain

$$\mathbf{f}_{2,1} = \mathbf{f}_{3,2} - m_2\mathbf{g} = \begin{bmatrix} f_x \\ f_y \\ f_z + m_3 g_c \end{bmatrix},$$

$$\begin{aligned} \mathbf{n}_{2,1} &= \mathbf{n}_{3,2} + \mathbf{r}_2 \times \mathbf{f}_{2,1} - \mathbf{r}_{c2} \times m_2\mathbf{g} \\ &= \begin{bmatrix} n_x + f_z(d_3 s\theta_1 s\theta_2 + d_2 c\theta_1) - f_y d_3 c\theta_2 + m_3 g_c(0.5 d_3 s\theta_1 s\theta_2 + d_2 c\theta_1) \\ n_y - f_z(d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) + f_x d_3 c\theta_2 - m_3 g_c(0.5 d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) \\ n_z + f_y(d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) - f_x(d_3 s\theta_1 s\theta_2 + d_2 c\theta_1) \end{bmatrix}. \end{aligned}$$

For $i = 1$, we substitute $\mathbf{f}_{2,1}$ and $\mathbf{n}_{2,1}$ obtained in the preceding step along with $\mathbf{r}_1$ and $\mathbf{r}_{c1}$ into Eqs. (6.4) and (6.5). This produces

$$\mathbf{f}_{1,0} = \mathbf{f}_{2,1} - m_1\mathbf{g} = \begin{bmatrix} f_x \\ f_y \\ f_z + m_3 g_c \end{bmatrix},$$

$$\begin{aligned} \mathbf{n}_{1,0} &= \mathbf{n}_{2,1} + \mathbf{r}_1 \times \mathbf{f}_{1,0} - \mathbf{r}_{c1} \times m_1\mathbf{g} \\ &= \begin{bmatrix} n_x + f_z(d_3 s\theta_1 s\theta_2 + d_2 c\theta_1) - f_y d_3 c\theta_2 + m_3 g_c(0.5 d_3 s\theta_1 s\theta_2 + d_2 c\theta_1) \\ n_y - f_z(d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) + f_x d_3 c\theta_2 - m_3 g_c(0.5 d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) \\ n_z + f_y(d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) - f_x(d_3 s\theta_1 s\theta_2 + d_2 c\theta_1) \end{bmatrix}. \end{aligned}$$

Finally, the joint torques are obtained by substituting the reaction forces above into Eqs. (6.16) and (6.17):

$$
\begin{aligned}
\tau_1 &= \mathbf{z}_0^{\mathrm{T}}\mathbf{n}_{1,0} = n_z + f_y(d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) - f_x(d_3 s\theta_1 s\theta_2 + d_2 c\theta_1),\\
\tau_2 &= \mathbf{z}_1^{\mathrm{T}}\mathbf{n}_{2,1}\\
&= -s\theta_1[n_x + f_z(d_3 s\theta_1 s\theta_2 + d_2 c\theta_1) - f_y d_3 c\theta_2\\
&\qquad + m_3 g_c(0.5 d_3 s\theta_1 s\theta_2 + d_2 c\theta_1)]\\
&\quad + c\theta_1[n_y - f_z(d_3 c\theta_1 s\theta_2 - d_2 s\theta_1) + f_x d_3 c\theta_2\\
&\qquad - m_3 g_c(0.5 d_3 c\theta_1 s\theta_2 - d_2 s\theta_1)],\\
\tau_3 &= \mathbf{z}_2^{\mathrm{T}}\mathbf{f}_{3,2} = f_x c\theta_1 s\theta_2 + f_y s\theta_1 s\theta_2 + (f_z + m_3 g_c) c\theta_2,\\
\tau_4 &= \mathbf{z}_3^{\mathrm{T}}\mathbf{n}_{4,3} = n_x c\theta_1 s\theta_2 + n_y s\theta_1 s\theta_2 + n_z c\theta_2,\\
\tau_5 &= \mathbf{z}_4^{\mathrm{T}}\mathbf{n}_{5,4} = n_x j_{45} + n_y j_{55} + n_z j_{65},\\
\tau_6 &= \mathbf{z}_5^{\mathrm{T}}\mathbf{n}_{6,5} = n_x j_{46} + n_y j_{56} + n_z j_{66},
\end{aligned}
$$

where j_{45}, j_{55}, and j_{65} are the x, y, and z components of the unit vector $\mathbf{z}_4$, and j_{46}, j_{56}, and j_{66} are the x, y, and z components of $\mathbf{z}_5$.

In the absence of gravity, the input joint torques and the output forces are related by the following transformation:

$$
\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \\ \tau_5 \\ \tau_6 \end{bmatrix} = J^{\mathrm{T}} \begin{bmatrix} f_x \\ f_y \\ f_z \\ n_x \\ n_y \\ n_z \end{bmatrix}, \tag{6.20}
$$

where

$$
J = \begin{bmatrix}
-d_3 s\theta_1 s\theta_2 - d_2 c\theta_1 & d_3 c\theta_1 c\theta_2 & c\theta_1 s\theta_2 & 0 & 0 & 0\\
d_3 c\theta_1 s\theta_2 - d_2 s\theta_1 & d_3 s\theta_1 c\theta_2 & s\theta_1 s\theta_2 & 0 & 0 & 0\\
0 & -d_3 s\theta_2 & c\theta_2 & 0 & 0 & 0\\
0 & -s\theta_1 & 0 & c\theta_1 s\theta_2 & j_{45} & j_{46}\\
0 & c\theta_1 & 0 & s\theta_1 s\theta_2 & j_{55} & j_{56}\\
1 & 0 & 0 & c\theta_2 & j_{65} & j_{66}
\end{bmatrix}
$$

is the conventional Jacobian matrix derived in Chapter 4.

From the two examples above, we observe that by neglecting the gravitational effect, the end-effector output forces are related to the input joint torques by the transpose of the conventional Jacobian matrix. In the following section we show that this relation holds for all serial manipulators.

6.2.4 Application of the Principle of Virtual Work

In this section we apply the principle of virtual work to derive a transformation between the joint torques and end-effector forces (Asada and Slotine, 1986; Paul, 1981; Witney, 1972). A virtual displacement of a system refers to an infinitesimal change in the configuration of the system as a result of any arbitrary infinitesimal changes of the coordinates that are compatible with the forces and constraints imposed on the system at a given instant in time. The term *virtual displacement* is used to distinguish it from an *actual displacement*, for which the forces and constraints may be changing at the instant. We use the Greek letter delta, $\delta\mathbf{x}$, to denote a virtual displacement, as opposed to $d\mathbf{x}$ for an actual displacement.

For a serial manipulator, the virtual displacements at the joints can be written as $\delta\mathbf{q} = [\delta q_1, \delta q_2, \ldots, \delta q_n]^{\mathrm{T}}$, and the virtual displacement of the end effector can be expressed as $\delta\mathbf{x} = [\delta x, \delta y, \ldots, \delta\psi]^{\mathrm{T}}$. Let the end-effector output force and moment be denoted by

$$\mathbf{F} = \begin{bmatrix} \mathbf{f} \\ \mathbf{n} \end{bmatrix}. \tag{6.21}$$

Also let the vector of joint torques be denoted by

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \vdots \\ \tau_n \end{bmatrix}. \tag{6.22}$$

Assuming that frictional forces at the joints are negligible, the virtual work produced by the forces of constraint at the joints is zero. Hence, by neglecting the gravitational effect, the virtual work, δW, done by all the active forces is given by

$$\delta W = \boldsymbol{\tau}^{\mathrm{T}}\delta\mathbf{q} - \mathbf{F}^{\mathrm{T}}\delta\mathbf{x}. \tag{6.23}$$

The principle of virtual work states that a system is under equilibrium if and only if the virtual work vanishes for any infinitesimal virtual displacement. This is true if the virtual displacements are compatible with the constraints imposed on the system. In Eq. (6.23), however, the virtual displacements $\delta\mathbf{q}$ and $\delta\mathbf{x}$ are not independent. In fact, they are related by the conventional Jacobian matrix as follows:

$$\delta \mathbf{x} = J\, \delta \mathbf{q}. \tag{6.24}$$

Substituting Eq. (6.24) into (6.23) yields

$$(\boldsymbol{\tau}^{\mathrm{T}} - \mathbf{F}^{\mathrm{T}} J)\, \delta \mathbf{q} = 0. \tag{6.25}$$

Since Eq. (6.25) holds for any arbitrary virtual displacement, $\delta \mathbf{q}$, we conclude that

$$\boldsymbol{\tau}^{\mathrm{T}} - \mathbf{F}^{\mathrm{T}} J = 0. \tag{6.26}$$

Taking the transpose of Eq. (6.26) yields

$$\boldsymbol{\tau} = J^{\mathrm{T}} \mathbf{F}. \tag{6.27}$$

Equation (6.27) maps an m-dimensional end-effector output force into an n-dimensional joint torques. Since the Jacobian matrix is configuration dependent, the mapping is also configuration dependent.

6.2.5 Force Ellipsoid

Similar to the transformation of velocities, the transformation of forces for manipulators with only one type of joints and for one type of tasks can be characterized by a comparison of the end-effector force produced by a unit joint torque. Substituting Eq. (6.27) into $\boldsymbol{\tau}^{\mathrm{T}} \boldsymbol{\tau} = 1$ yields

$$\mathbf{F}^{\mathrm{T}} J J^{\mathrm{T}} \mathbf{F} = 1. \tag{6.28}$$

At a given manipulator configuration, Eq. (6.28) represents an m-dimensional ellipsoid. Because the product $J J^{\mathrm{T}}$ is symmetric positive semidefinite, its eigenvectors are orthogonal. The principal axes of the ellipsoid coincide with the eigenvectors of $J J^{\mathrm{T}}$, and their lengths are equal to the reciprocals of the square roots of the eigenvalues.

Since the Jacobian matrix is configuration dependent, the force ellipsoid is also configuration dependent. As the end effector moves from one location to another, the shape and orientation of the force ellipsoid will also change accordingly. The closer the transmission ellipsoid to a sphere, the better the transmission characteristics are. The transformation is said to be *isotropic* when the principal axes are of equal lengths. At an isotropic point, an n-dimensional unit sphere in the joint torque space maps onto an m-dimensional sphere in the end-effector force space. On the other hand, at a singular point, an n-dimensional unit sphere in the joint torque space maps onto an m-

dimensional cylinder in the end-effector force space. Thus the mechanical advantage of the manipulator becomes infinitely large in some direction.

Example 6.2.3 *Ellipsoid of a Planar 2-DOF Manipulator* In this example we examine the transformation characteristics of the planar 2-dof manipulator shown in Fig. 6.4 to illustrate the principle. For this 2-dof manipulator, the end-effector output force and input joint torques can be written as $\mathbf{f} = [f_x, f_y]^{\mathrm{T}}$ and $\tau = [\tau_x, \tau_y]^{\mathrm{T}}$, respectively.

Substituting the Jacobian matrix, Eq. (4.66), into (6.27), we obtain

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} -a_1 s\theta_1 - a_2 s\theta_{12} & a_1 c\theta_1 + a_2 c\theta_{12} \\ -a_2 s\theta_{12} & a_2 c\theta_{12} \end{bmatrix} \begin{bmatrix} f_x \\ f_y \end{bmatrix}. \tag{6.29}$$

Let the link lengths be $a_1 = \sqrt{2}$ m and $a_2 = 1$ m. At the posture where $\theta_1 = 0$ and $\theta_2 = \pi/2$, the Jacobian matrix reduces to

$$J = \begin{bmatrix} -1 & -1 \\ \sqrt{2} & 0 \end{bmatrix}.$$

Hence

$$J J^{\mathrm{T}} = \begin{bmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix}.$$

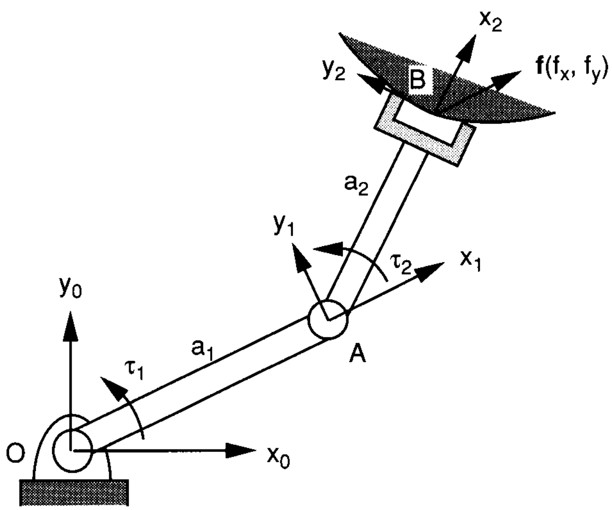

FIGURE 6.4. Planar $2R$ manipulator exerting a force $\mathbf{f}(f_x, f_y)$.

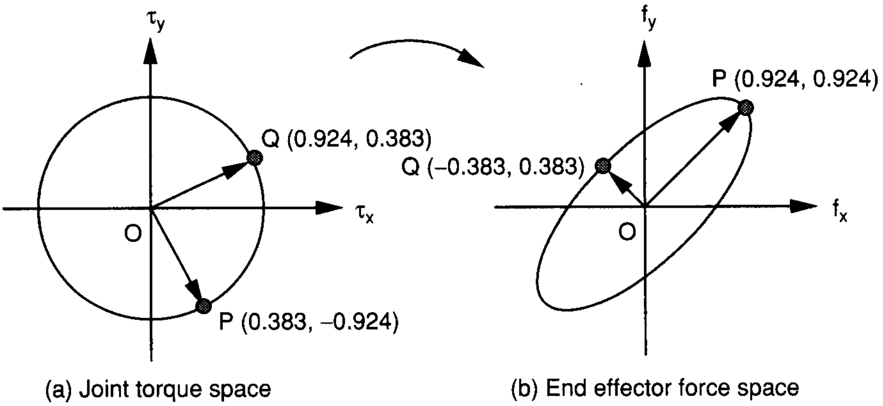

FIGURE 6.5. Force ellipsoid.

The eigenvalues of JJ^{T} are $\lambda_1 = 2 - \sqrt{2} = 0.5858$ and $\lambda_2 = 2 + \sqrt{2} = 3.4142$. The corresponding eigenvectors, normalized to unit length, are (0.707, 0.707) and (-0.707, 0.707), respectively. These two eigenvectors are at 45° angles with the f_x and f_y axes, respectively, and they are lined up with the principal axes of the ellipse.

Substituting JJ^{T} into (6.28), we obtain

$$2f_x^2 - 2\sqrt{2}f_x f_y + 2f_y^2 = 0.5858\left(\frac{f_x}{\sqrt{2}} + \frac{f_y}{\sqrt{2}}\right)^2 + 3.4142\left(\frac{f_x}{\sqrt{2}} - \frac{f_y}{\sqrt{2}}\right)^2$$
$$= 1.$$

Figure 6.5 shows the ellipse and its principal axes. The end-effector forces produced by a unit joint torque are $(f_x, f_y) = (0.924, 0.924)$ N along the major axis and $(f_x, f_y) = (-0.383, 0.383)$ N along the minor axis. The corresponding joint torques are $(\tau_1, \tau_2) = (0.383, -0.924)$ N·m along the major axis and $(\tau_1, \tau_2) = (0.924, 0.383)$ N·m along the minor axis. We note that the mechanical advantage along the major axis is larger than that along the minor axis.

6.3 TRANSFORMATION OF FORCES AND MOMENTS

In Chapter 4, we have shown that the coordinates of a twist can be transformed from one coordinate frame to another by a 6×6 transformation matrix ${}^i\tilde{T}_j$. Analogously, the coordinates of a wrench can be transformed from one coordinate frame to another by the same matrix. Let **f** denote the force acting along, and **c** denote the couple about, the axis of a wrench. Then the wrench can be written as a six-dimensional column vector:

$$\mathbf{F} = \begin{bmatrix} \mathbf{f} \\ \mathbf{s}_o \times \mathbf{f} + \mathbf{c} \end{bmatrix} = \begin{bmatrix} f_x \\ f_y \\ f_z \\ n_x \\ n_y \\ n_z \end{bmatrix}, \tag{6.30}$$

where $\mathbf{s}_o$ denotes the position vector of any point on the wrench axis relative to a reference frame. The vector $\mathbf{s}_o \times \mathbf{f}$ denotes the moment contributed by the force **f** about the origin of the reference frame chosen. We observe that the coordinates of a wrench depend on the choice of the reference frame, due to the fact that the effect of a force on the resulting mement, $\mathbf{n} = [n_x, n_y, n_z]^T$, depends on the point of application.

Figure 6.6 shows a wrench **F** with reference to two reference frames: (x_i, y_i, z_i) and (x_j, y_j, z_j). The position of the (x_j, y_j, z_j) frame relative to the (x_i, y_i, z_i) frame is denoted by a position vector ${}^i\mathbf{p} = [p_x, p_y, p_z]^T$, and the orientation of the (x_j, y_j, z_j) frame relative to the (x_i, y_i, z_i) frame is described by a rotation matrix ${}^iR_j$. The wrench **F** relative to the (x_i, y_i, z_i) frame is denoted by ${}^i\mathbf{F}$, and the same wrench relative to the (x_j, y_j, z_j) frame

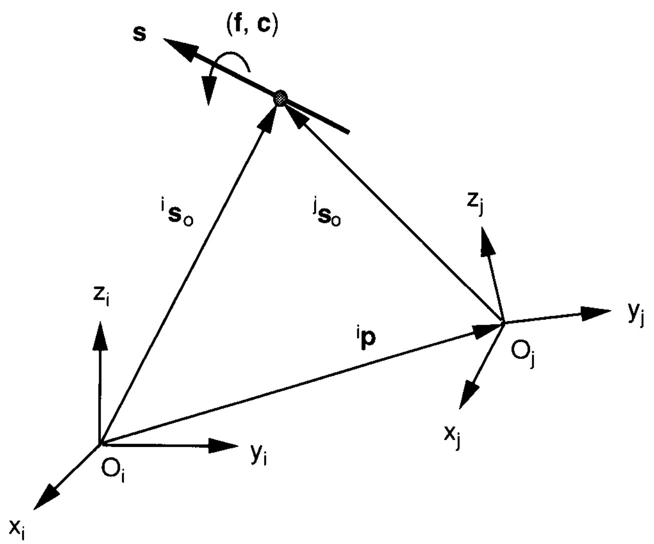

FIGURE 6.6. Transformation of force and moment between two coordinate systems.

is denoted by ${}^j\mathbf{F}$. Since the wrench has the properties of a screw, it follows that

$$ {}^i\mathbf{F} = {}^i\tilde{T}_j \, {}^j\mathbf{F}, \tag{6.31} $$

where ${}^i\tilde{T}_j$ is the 6×6 matrix derived in Chapter 4. Specifically,

$$ {}^i\tilde{T}_j = \left[\begin{array}{c:c} {}^iR_j & \mathbf{0} \\ \hdashline {}^iW_j \, {}^iR_j & {}^iR_j \end{array} \right], \tag{6.32} $$

where

$$ {}^iW_j = \begin{bmatrix} 0 & -p_z & p_y \\ p_z & 0 & -p_x \\ -p_y & p_x & 0 \end{bmatrix} $$

is a 3×3 skew-symmetric matrix representing the vector of $\overline{O_i O_j}$ expressed in the (x_i, y_i, z_i) frame. Hence, given a wrench about one reference frame, we can transform it into another reference frame, and vice versa.

Example 6.3.1 *Transformation between Tool Tip and Wrist Center* Figure 6.7 shows a robot hand inserting a peg in a hole. We wish to monitor the force and moment acting on the peg. To accomplish this, a force sensor is mounted at the wrist center. Thus the force and moment are referred to two different locations: one with respect to the (x, y, z) frame at the peg and the other with respect to the (u, v, w) frame at the wrist center. The force and moment with respect to the (u, v, w) coordinate frame are known from the measurement made at the wrist center. The problem is to find the force and moment acting on the peg.

Assuming that these two coordinate frames are parallel at the instant and the wrist center Q relative to the peg coordinate frame is given by the vector $\mathbf{p} = [p_x, p_y, p_z]^T$, we have

$$ {}^{xyz}R_{uvw} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, $$

$$ {}^{xyz}W_{uvw} = \begin{bmatrix} 0 & -p_z & p_y \\ p_z & 0 & -p_x \\ -p_y & p_x & 0 \end{bmatrix}. $$

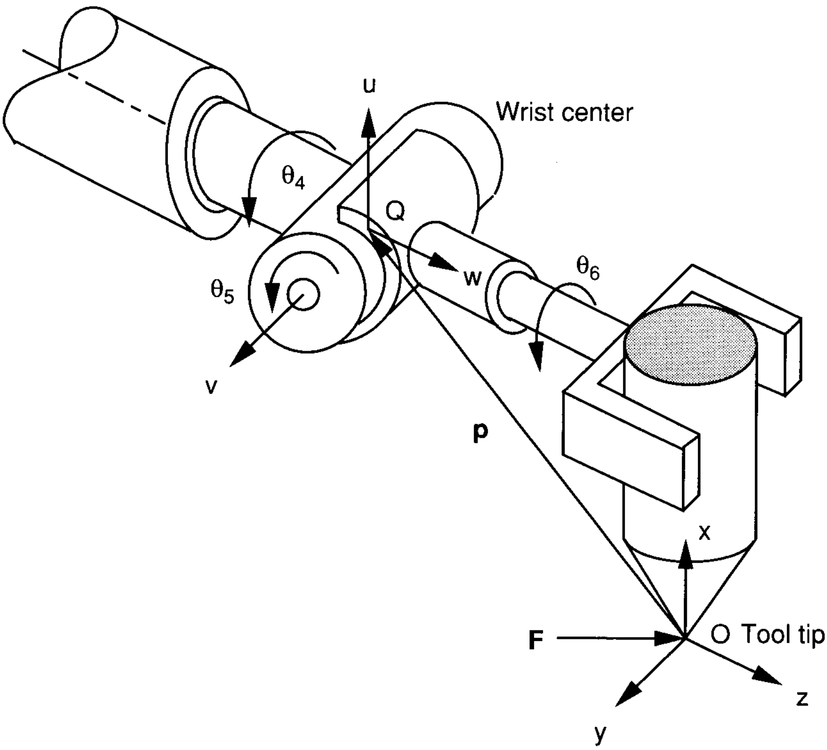

FIGURE 6.7. Transformation of force and moment from the tool tip to the wrist center.

Hence the transformation matrix ${}^{xyz}\tilde{T}_{uvw}$ is given by

$$
{}^{xyz}\tilde{T}_{uvw} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -p_z & p_y & 1 & 0 & 0 \\ p_z & 0 & -p_x & 0 & 1 & 0 \\ -p_y & p_x & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{6.33}
$$

Substituting Eq. (6.33) into Eq. (6.31), we obtain

$$
\begin{bmatrix} f_x \\ f_y \\ f_z \\ n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -p_z & p_y & 1 & 0 & 0 \\ p_z & 0 & -p_x & 0 & 1 & 0 \\ -p_y & p_x & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f_u \\ f_v \\ f_w \\ n_u \\ n_v \\ n_w \end{bmatrix}. \tag{6.34}
$$

Equation (6.34) provides a transformation of the force and moment measured at the wrist center to the output force and moment at the inserting point.

6.4 STIFFNESS ANALYSIS OF SERIAL MANIPULATORS

When a manipulator performs a given task, the end effector exerts some force and/or moment on its environment. This contact force and/or moment will cause the end effector to be deflected away from its desired location. Intuitively, the amount of deflection is a function of the applied force and stiffness of the manipulator. Thus the stiffness of a manipulator has a direct impact on the position accuracy. Furthermore, some advanced control strategies make use of the stiffness characteristics for feedback control of a robot manipulator (Bryfogle et al., 1993; Lebret et al., 1993).

The overall stiffness of a manipulator depends on several factors, including the size of and material used for the links, the mechanical transmission mechanisms, the actuators, and the controller. As the links become longer and more slender, link compliance becomes the major source of deflection. This is particularly true for space robots, for which light weight and compactness are the major concern (Nguyen and Ravindran, 1977). A few studies on the dynamics and control of flexible manipulators can be found in Cannon and Schmitz (1984), Chang and Hamilton (1991), and Hollars and Cannon (1985). On the other hand, most industrial robots are constructed with fairly rigid links, and the major sources of compliance come from the mechanical transmission mechanisms and control systems (Sunada and Dubowsky, 1983; Sweet and Good, 1984).

In what follows, we assume that the major links are perfectly rigid and consider the mechanical transmission mechanisms and the servo systems as the main sources of compliance.

6.4.1 Compliance Matrix

For a serial manipulator, each joint is typically driven by an actuator through a multiple-stage speed reducer along with several drive shafts. The speed reducer and the drive shafts may deflect when torque or force is transmitted. Further, the drive torque or force generated by a servo system usually depends on the position and velocity error signals and its feedback gains. The stiffness of the speed reducer, the drive shafts, and the servo system may be combined into an equivalent stiffness.

Let n be the dimension of the joint space, m be the dimension of the end-effector space, τ_i be the torque or force transmitted through the ith joint, and Δq_i be the corresponding deflection at the joint. Then for small deflections we can relate τ_i and Δq_i by the linear approximation

$$\tau_i = k_i \, \Delta q_i, \tag{6.35}$$

where k_i is called the *stiffness constant*. For convenience, we write Eq. (6.35) for $i = 1, 2, \ldots, n$ in matrix form:

$$\boldsymbol{\tau} = \chi \, \Delta \mathbf{q}, \tag{6.36}$$

where $\boldsymbol{\tau} = [\tau_1, \tau_2, \ldots, \tau_n]^{\mathrm{T}}$, $\Delta \mathbf{q} = [\Delta q_1, \Delta q_2, \ldots, \Delta q_n]^{\mathrm{T}}$, and $\chi = \mathrm{diag}[k_1, k_2, \ldots, k_n]$ is an $n \times n$ diagonal matrix.

For a serial manipulator, the joint displacement $\Delta \mathbf{q}$ is related to the end effector displacement $\Delta \mathbf{x}$ by the conventional $m \times n$ Jacobian matrix,

$$\Delta \mathbf{x} = J \Delta \mathbf{q}, \tag{6.37}$$

and the end-effector output force $\mathbf{F}$ is related to the joint torque $\boldsymbol{\tau}$ by the transpose of the conventional Jacobian matrix,

$$\boldsymbol{\tau} = J^{\mathrm{T}} \mathbf{F}. \tag{6.38}$$

Eliminating $\boldsymbol{\tau}$ and $\Delta \mathbf{q}$ from Eqs. (6.36), (6.37), and (6.38), we obtain

$$\Delta \mathbf{x} = C \mathbf{F}, \tag{6.39}$$

where

$$C = J \chi^{-1} J^{\mathrm{T}} \tag{6.40}$$

is an $m \times m$ matrix called the *compliance matrix*.

From Eq. (6.40) we note that the compliance matrix is symmetric. It depends not only on the stiffness of each drive line but also on the Jacobian matrix. Since the Jacobian matrix is configuration dependent, the compliance matrix is also configuration dependent.

6.4.2 Stiffness Matrix

If $m = n$ and the Jacobian matrix is nonsingular, the compliance matrix is invertible. Multiplying both sides of Eq. (6.39) by C^{-1}, we obtain

$$\mathbf{F} = K \, \Delta \mathbf{x}, \tag{6.41}$$

where

$$K = C^{-1} = J^{-\mathrm{T}} \chi J^{-1} \tag{6.42}$$

is called the *stiffness matrix.*

Obviously, the stiffness matrix is also configuration dependent. For manipulators with only one type of joint and for one type of task, we may characterize the stiffness of a manipulator by comparing the force required to generate one unit-length deflection at the end effector. Substituting Eq. (6.39) into $(\Delta\mathbf{x})^{\mathrm{T}}(\Delta\mathbf{x}) = 1$, we obtain

$$\mathbf{F}^{\mathrm{T}} C^{\mathrm{T}} C \mathbf{F} = 1. \tag{6.43}$$

At a given manipulator configuration, Eq. (6.43) represents an m-dimensional force ellipsoid. Because $C^{\mathrm{T}}C$ is symmetric positive semidefinite, its eigenvectors are orthogonal. The principal axes of the ellipsoid coincide with the eigenvectors of $C^{\mathrm{T}}C$, and their lengths are equal to the reciprocals of the square roots of the eigenvalues. Hence the maximum and minimum forces required to produce a unit deflection are given by $1/\sqrt{\lambda_{\min}}$ and $1/\sqrt{\lambda_{\max}}$, respectively, where $\lambda_{\min}$ and $\lambda_{\max}$ are the minumum and maximum eigenvalues of $C^{\mathrm{T}}C$.

Example 6.4.1 *Stiffness Analysis of a Planar 2-DOF Manipulator* Consider the planar 2-dof manipulator shown in Fig. 6.4 as an example. The end-effector deflection and output force are $\Delta\mathbf{x} = [\Delta x, \Delta y]^{\mathrm{T}}$ and $\mathbf{F} = [f_x, f_y]^{\mathrm{T}}$, respectively. We wish to examine the stiffness properties of this manipulator. Substituting the Jacobian matrix, Eq. (4.66), into (6.40), we obtain

$$C = \begin{bmatrix} \dfrac{(a_1 s_1 + a_2 s_{12})^2}{k_1} + \dfrac{a_2^2 s_{12}^2}{k_2} & -\dfrac{(a_1 s_1 + a_2 s_{12})(a_1 c_1 + a_2 c_{12})}{k_1} - \dfrac{a_2^2 s_{12} c_{12}}{k_2} \\ -\dfrac{(a_1 s_1 + a_2 s_{12})(a_1 c_1 + a_2 c_{12})}{k_1} - \dfrac{a_2^2 s_{12} c_{12}}{k_2} & \dfrac{(a_1 c_1 + a_2 c_{12})^2}{k_1} + \dfrac{a_2^2 c_{12}^2}{k_2} \end{bmatrix}, \tag{6.44}$$

where $s_i = \sin\theta_i$, $c_i = \cos\theta_i$, $s_{ij} = \sin(\theta_i + \theta_j)$, and $c_{ij} = \cos(\theta_i + \theta_j)$.

For the purpose of demonstration, let $a_1 = \sqrt{2}$ m, $a_2 = 1$ m, $k_1 = k_2 = 1\,\mathrm{N{\cdot}m}$, $\theta_1 = 0$, and $\theta_2 = \pi/2$. Then the compliance matrix reduces to

$$C = \begin{bmatrix} 2 & -\sqrt{2} \\ -\sqrt{2} & 2 \end{bmatrix}.$$

Hence

$$C^{\mathrm{T}}C = \begin{bmatrix} 6 & -4\sqrt{2} \\ -4\sqrt{2} & 6 \end{bmatrix}.$$

Equation (6.43) reduces to

$$6f_x^2 - 8\sqrt{2}\, f_x f_y + 6f_y^2 = 0.343\left(\frac{f_x}{\sqrt{2}} + \frac{f_y}{\sqrt{2}}\right)^2 + 11.657\left(\frac{f_x}{\sqrt{2}} - \frac{f_y}{\sqrt{2}}\right)^2 = 1.$$

The equation above represents an ellipse in the end-effector force space. The eigenvalues of $C^{\mathrm{T}}C$ are $\lambda_1 = 0.343$ and $\lambda_2 = 11.657$. The corresponding eigenvectors, normalized to unit length, are (0.707, 0.707) and (−0.707, 0.707), respectively. These two eigenvectors are at 45° angles with the f_x and f_y axes, respectively, and they line up with the principal axes of the ellipse.

6.5 STATICS OF PARALLEL MANIPULATORS

In this section we study the statics of parallel manipulators, which is more complicated than that of its serial counterpart. Due to the existence of several closed loops, the recursive method of analysis is no longer applicable. In general, it is necessary to derive the force and moment balance equations for each link and solve the equations simultaneously. However, if only the actuator drive forces and/or torques are of interest, the principle of virtual work can be applied (Bryfogle et al., 1993; Gosselin, 1990; Tahmasebi and Tsai, 1995).

6.5.1 Free-Body Diagram Approach

We first investigate the free-body diagram approach. A parallel manipulator typically consists of several limbs, each made up of several binary links. The analysis can be simplified greatly by making use of the fact that some of these binary links will bear only tension or compression forces. For example, if no external force or moment is applied to the binary link of a planar mechanism, the link will only be subject to tension or compression force along a line passing through the two revolute joints. Similarly, if no external force or moment is applied to a spherical–spherical binary link chain in a spatial mechanism, the link will only be subject to tension or compression force along a line passing through the two spherical centers. This method can best be demonstrated by examples.

Example 6.5.1 *Statics of a Planar 3RRR Manipulator* In this example we study the statics of the planar 3-dof parallel manipulator shown in Fig. 5.2. There are three limbs, each consisting of two members. The first member, a_i, is the input link; the second member, b_i, serves as a coupling link. Assuming that the acceleration of gravity is perpendicular to the plane of motion, we wish to find the joint torques, τ_1, τ_2, and τ_3, required at the input links to produce an output force $\mathbf{f} = [f_x, f_y, 0]^T$ and an output moment $\mathbf{n} = [0, 0, n_z]^T$ at point G.

The forces exerted at the coupler link pass through its two end pivots, which can be written as

$$\mathbf{f}_i = \frac{f_i \mathbf{b}_i}{b_i} \qquad \text{for } i = 1, 2, 3, \tag{6.45}$$

where the subscript i denotes the ith limb, f_i denotes the magnitude of $\mathbf{f}_i$, $\mathbf{b}_i$ for $i = 1$ to 3 denote the vectors $\overline{DA}$, $\overline{EB}$, and $\overline{FC}$, respectively, and b_i denotes the magnitude of $\mathbf{b}_i$.

Summing all the forces acting on the moving platform, we obtain

$$\sum_{i=1}^{3} \frac{f_i \mathbf{b}_i}{b_i} = [f_x, f_y, 0]^T. \tag{6.46}$$

Summing the moments contributed by all forces about G, we obtain

$$\sum_{i=1}^{3} \frac{\mathbf{e}_i \times f_i \mathbf{b}_i}{b_i} = [0, 0, n_z]^T. \tag{6.47}$$

Equations (6.46) and (6.47) contain three nontrivial scalar equations in three unknowns, f_i for $i = 1, 2$, and 3. These three linear equations can be written in matrix form as

$$\begin{bmatrix} \frac{b_{1x}}{b_1} & \frac{b_{2x}}{b_2} & \frac{b_{3x}}{b_3} \\ \frac{b_{1y}}{b_1} & \frac{b_{2y}}{b_2} & \frac{b_{3y}}{b_3} \\ \frac{e_{1x}b_{1y} - e_{1y}b_{1x}}{b_1} & \frac{e_{2x}b_{2y} - e_{2y}b_{2x}}{b_2} & \frac{e_{3x}b_{3y} - e_{3y}b_{3x}}{b_3} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} f_x \\ f_y \\ n_z \end{bmatrix}. \tag{6.48}$$

The vectors $\mathbf{e}_i$ and $\mathbf{b}_i$ are known from the kinematic analysis. Hence the three unknown forces, f_i for $i = 1, 2$, and 3, can be found from the inverse transformation of Eq. (6.48). Once f_i for $i = 1, 2$, and 3 are known, the actuator torques can be found from the moment balance equation of each input link about its fixed pivot; that is,

$$\tau_i = \frac{f_i(a_{ix}b_{iy} - a_{iy}b_{ix})}{b_i}. \tag{6.49}$$

Eliminating f_i between Eqs. (6.48) and (6.49), we obtain

$$\begin{bmatrix} f_x \\ f_y \\ n_z \end{bmatrix} = \begin{bmatrix} \dfrac{b_{1x}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \dfrac{b_{2x}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \dfrac{b_{3x}}{a_{3x}b_{3y} - a_{3y}b_{3x}} \\ \dfrac{b_{1y}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \dfrac{b_{2y}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \dfrac{b_{3y}}{a_{3x}b_{3y} - a_{3y}b_{3x}} \\ \dfrac{e_{1x}b_{1y} - e_{1y}b_{1x}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \dfrac{e_{2x}b_{2y} - e_{2y}b_{2x}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \dfrac{e_{3x}b_{3y} - e_{3y}b_{3x}}{a_{3x}b_{3y} - a_{3y}b_{3x}} \end{bmatrix} \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}. \tag{6.50}$$

Equation (6.50) provides a transformation between the end-effector output forces and the input joint torques. Given the input joint torques, the end-effector output forces can be found directly from Eq. (6.50). On the other hand, given the end-effector output forces, we can find the input joint torques by the inverse transformation of Eq. (6.50).

Example 6.5.2 *Statics of the Stewart–Gough Platform* Consider the Stewart–Gough platform shown in Fig. 3.15. Let the origin O of the fixed (x, y, z) frame be located at the centroid of the fixed base with the x and y axes lying on the plane of the attachment points, A_i, for $i = 1$ to 6. Also let the origin P of the moving (u, v, w) frame be located at the centroid of the moving platform, with its u and v axes lying on the plane of the attachment points, B_i, for $i = 1$ to 6. Furthermore, let the gravitational effects be negligible. We wish to find the actuator forces, $f_1, f_2, \ldots, f_6$, required to produce an output force $\mathbf{f}$ and an output moment $\mathbf{n}$ at the centroid of the moving platform.

The forces and moments of constraint associated each prismatic joint can be considered as internal to the limb. Because of the spherical–spherical limb construction, no moment can be transmitted to the limb. In addition, the reaction force at each spherical joint points along line A_iB_i defined by the two spherical centers. Hence the force acting on the moving platform by each limb can be written as

$$\mathbf{f}_i = f_i\mathbf{s}_i \qquad \text{for } i = 1, 2, \ldots, 6, \tag{6.51}$$

where the subscript i denotes the ith limb, f_i denotes the magnitude of $\mathbf{f}_i$, and $\mathbf{s}_i$ denotes a unit vector pointing from the base spherical joint to the moving spherical joint of the ith limb, namely, $\mathbf{s}_i = \mathbf{d}_i / d_i$.

We now form the force and moment balance equations of the moving platform. Summing all the forces acting on the moving platform, we obtain

$$\sum_{i=1}^{6} f_i \mathbf{s}_i = \mathbf{f}. \tag{6.52}$$

Summing the moments contributed by all forces about the centroid P of the moving platform, we obtain

$$\sum_{i=1}^{6} f_i \mathbf{b}_i \times \mathbf{s}_i = \mathbf{n}, \tag{6.53}$$

where $\mathbf{b}_i = \overline{PB_i}$. Equations (6.52) and (6.53) represent six linear scalar equations in f_i which can be written in matrix form as

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{n} \end{bmatrix} = \begin{bmatrix} \mathbf{s}_1 & \mathbf{s}_2 & \cdots & \mathbf{s}_6 \\ \mathbf{b}_1 \times \mathbf{s}_1 & \mathbf{b}_2 \times \mathbf{s}_2 & \cdots & \mathbf{b}_6 \times \mathbf{s}_6 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_6 \end{bmatrix}. \tag{6.54}$$

Equation (6.54) provides a transformation between the end-effector output forces and the linear actuator forces. The vectors $\mathbf{b}_i$ and $\mathbf{s}_i$ are known from the kinematic analysis. Hence given the actuator forces, the end-effector output forces can be computed directly from Eq. (6.54). On the other hand, if the end-effector output forces are given, we can find the corresponding actuator forces by the inverse transformation of Eq. (6.54).

6.5.2 Application of the Principle of Virtual Work

As mentioned in Chapter 5, there are two types of joints in a parallel manipulator: actuated joints and passive joints. Assuming that the joints are frictionless and the gravitational effects are negligible, the reaction forces at the passive joints contribute to no virtual work.

Let $\mathbf{F} = [\mathbf{f}, \mathbf{n}]^T$ represent the vector of end-effector output force and moment, $\boldsymbol{\tau} = [\tau_1, \tau_2, \ldots, \tau_n]^T$ represent the vector of actuated joint torques or forces, $\delta\mathbf{q} = [\delta q_1, \delta q_2, \ldots, \delta q_n]^T$ represent the vector of virtual displacements associated with the actuated joints, and $\delta\mathbf{x} = [\delta x, \delta y, \ldots, \delta\psi]^T$ represent the vector of virtual displacements associated with the end effector. Then the virtual work contributed by all the active forces can be written as

$$\boldsymbol{\tau}^{\mathrm{T}}\delta\mathbf{q} - \mathbf{F}^{\mathrm{T}}\delta\mathbf{x} = 0. \tag{6.55}$$

However, the virtual displacements, $\delta\mathbf{q}$ and $\delta\mathbf{x}$, are related by the Jacobian matrix

$$\delta\mathbf{q} = J\,\delta\mathbf{x}, \tag{6.56}$$

where $J = J_q^{-1}J_x$ is as defined in Chapter 5.

Substituting Eq. (6.56) into (6.55) yields

$$(\boldsymbol{\tau}^{\mathrm{T}}J - \mathbf{F}^{\mathrm{T}})\,\delta\mathbf{x} = 0. \tag{6.57}$$

Since Eq. (6.57) holds for any virtual displacement, $\delta\mathbf{x}$, we conclude that

$$\boldsymbol{\tau}^{\mathrm{T}}J - \mathbf{F}^{\mathrm{T}} = 0. \tag{6.58}$$

Taking the transpose of Eq. (6.58) yields

$$\mathbf{F} = J^{\mathrm{T}}\boldsymbol{\tau}. \tag{6.59}$$

Equation (6.59) gives the end-effector output forces in terms of the actuated joint torques, and vice versa, via the inverse transformation. Because the Jacobian matrix for a parallel manipulator is defined as the inverse of that for a serial manipulator, Eq. (6.59) takes a different form from that of Eq. (6.27). Clearly, the transformation depends on the posture of a robot.

Example 6.5.3 *Planar 3RRR Parallel Manipulator* In this example we perform the static analysis of the planar 3-dof parallel manipulator shown in Fig. 5.2 by the principle of virtual work. The Jacobian of this manipulator was derived in Chapter 5. The vector of input joint rates is $\dot{\mathbf{q}} = [\dot\theta_1, \dot\theta_2, \dot\theta_3]^{\mathrm{T}}$, and the vector of end-effector velocities is $\dot{\mathbf{x}} = [v_{gx}, v_{gy}, \dot\phi]^{\mathrm{T}}$. The Jacobian matrices, derived in Chapter 5, are

$$J_x = \begin{bmatrix} b_{1x} & b_{1y} & e_{1x}b_{1y} - e_{1y}b_{1x} \\ b_{2x} & b_{2y} & e_{2x}b_{2y} - e_{2y}b_{2x} \\ b_{3x} & b_{3y} & e_{3x}b_{3y} - e_{3y}b_{3x} \end{bmatrix}, \tag{6.60}$$

$$J_q = \begin{bmatrix} a_{1x}b_{1y} - a_{1y}b_{1x} & 0 & 0 \\ 0 & a_{2x}b_{2y} - a_{2y}b_{2x} & 0 \\ 0 & 0 & a_{3x}b_{3y} - a_{3y}b_{3x} \end{bmatrix}. \tag{6.61}$$

Assuming that $\mathbf{a}_i \times \mathbf{b}_i \neq 0$, the overall Jacobian matrix is given by

$$J = J_q^{-1} J_x = \begin{bmatrix} \frac{b_{1x}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \frac{b_{1y}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \frac{e_{1x}b_{1y} - e_{1y}b_{1x}}{a_{1x}b_{1y} - a_{1y}b_{1x}} \\ \frac{b_{2x}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \frac{b_{2y}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \frac{e_{2x}b_{2y} - e_{2y}b_{2x}}{a_{2x}b_{2y} - a_{2y}b_{2x}} \\ \frac{b_{3x}}{a_{3x}b_{3y} - a_{3y}b_{3x}} & \frac{b_{3y}}{a_{3x}b_{3y} - a_{3y}b_{3x}} & \frac{e_{3x}b_{3y} - e_{3y}b_{3x}}{a_{3x}b_{3y} - a_{3y}b_{3x}} \end{bmatrix}. \tag{6.62}$$

Substituting Eq. (6.62) into (6.59), we obtain

$$\begin{bmatrix} f_x \\ f_y \\ n_z \end{bmatrix} = \begin{bmatrix} \frac{b_{1x}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \frac{b_{2x}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \frac{b_{3x}}{a_{3x}b_{3y} - a_{3y}b_{3x}} \\ \frac{b_{1y}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \frac{b_{2y}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \frac{b_{3y}}{a_{3x}b_{3y} - a_{3y}b_{3x}} \\ \frac{e_{1x}b_{1y} - e_{1y}b_{1x}}{a_{1x}b_{1y} - a_{1y}b_{1x}} & \frac{e_{2x}b_{2y} - e_{2y}b_{2x}}{a_{2x}b_{2y} - a_{2y}b_{2x}} & \frac{e_{3x}b_{3y} - e_{3y}b_{3x}}{a_{3x}b_{3y} - a_{3y}b_{3x}} \end{bmatrix} \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}. \tag{6.63}$$

Hence we have arrived at the same result as that derived by the free-body diagram analysis.

Example 6.5.4 *Stewart–Gough Platform* Let us consider the Stewart–Gough platform shown in Fig. 3.15 as a second example. Using the conventional approach, the end-effector velocity state is defined as

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{v}_p \\ \boldsymbol{\omega}_B \end{bmatrix}, \tag{6.64}$$

and the input joint rates are described by a six-dimensional vector,

$$\dot{\mathbf{q}} = [\dot{d}_1, \dot{d}_2, \ldots, \dot{d}_6]^{\mathrm{T}}. \tag{6.65}$$

The Jacobian matrices, derived in Chapter 5, are

$$J_x = \begin{bmatrix} \mathbf{s}_1^{\mathrm{T}} & (\mathbf{b}_1 \times \mathbf{s}_1)^{\mathrm{T}} \\ \mathbf{s}_2^{\mathrm{T}} & (\mathbf{b}_2 \times \mathbf{s}_2)^{\mathrm{T}} \\ \vdots & \vdots \\ \mathbf{s}_6^{\mathrm{T}} & (\mathbf{b}_6 \times \mathbf{s}_6)^{\mathrm{T}} \end{bmatrix} \tag{6.66}$$

$$J_q = I \tag{6.67}$$

Hence the overall Jacobian matrix is given by

$$J = J_q^{-1} J_x = \begin{bmatrix} \mathbf{s}_1^{\mathrm{T}} & (\mathbf{b}_1 \times \mathbf{s}_1)^{\mathrm{T}} \\ \mathbf{s}_2^{\mathrm{T}} & (\mathbf{b}_2 \times \mathbf{s}_2)^{\mathrm{T}} \\ \vdots & \vdots \\ \mathbf{s}_6^{\mathrm{T}} & (\mathbf{b}_6 \times \mathbf{s}_6)^{\mathrm{T}} \end{bmatrix}. \tag{6.68}$$

Substituting Eq. (6.68) into (6.59), we obtain

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{n} \end{bmatrix} = \begin{bmatrix} \mathbf{s}_1 & \mathbf{s}_2 & \cdots & \mathbf{s}_6 \\ \mathbf{b}_1 \times \mathbf{s}_1 & \mathbf{b}_2 \times \mathbf{s}_2 & \cdots & \mathbf{b}_6 \times \mathbf{s}_6 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_6 \end{bmatrix}. \tag{6.69}$$

Again, we have arrived at the same result as that derived by the free-body diagram analysis.

6.6 STIFFNESS ANALYSIS OF PARALLEL MANIPULATORS

The stiffness analysis of parallel manipulators has been the subject of recent study by several workers (Gosselin, 1990; Kerr, 1989; Lebret et al., 1993; Tahmasebi and Tsai, 1995). We shall assume that, similar to serial manipulators, the links are perfectly rigid and consider the mechanical transmission mechanisms and servo systems as the only sources of compliance.

Let $\boldsymbol{\tau} = [\tau_1, \tau_2, \ldots, \tau_n]^{\mathrm{T}}$ be the vector of actuated joint torques or forces and $\Delta\mathbf{q} = [\Delta q_1, \Delta q_2, \ldots, \Delta q_n]^{\mathrm{T}}$ be the corresponding vector of joint deflections. Then we can relate $\Delta\mathbf{q}$ to $\boldsymbol{\tau}$ by an $n \times n$ diagonal matrix, $\chi = \mathrm{diag}[k_1, k_2, \ldots, k_n]$, as follows:

$$\boldsymbol{\tau} = \chi\, \Delta\mathbf{q}. \tag{6.70}$$

For a parallel manipulator, the joint displacement, $\Delta\mathbf{q}$, is related to the end-effector deflection, $\Delta\mathbf{x} = [\Delta x, \Delta y, \Delta z, \Delta\phi, \Delta\theta, \Delta\psi]^{\mathrm{T}}$, by the Jacobian matrix, J:

$$\Delta\mathbf{q} = J\, \Delta\mathbf{x}, \tag{6.71}$$

where $J = J_q^{-1} J_x$ is the conventional Jacobian matrix defined in Chapter 5. Substituting Eq. (6.71) into (6.70) and the resulting equation into (6.59) yields

$$\mathbf{F} = K\, \Delta\mathbf{x}, \tag{6.72}$$

where

$$K = J^{\mathrm{T}} \chi J \tag{6.73}$$

is called the *stiffness matrix* of a parallel manipulator.

Equation (6.72) implies that the end-effector output force is related to its deflection by the stiffness matrix K. The stiffness matrix is symmetric, positive semidefinite, and manipulator configuration dependent. Furthermore, if the limbs are of the same type and the spring constants associated with all the drive lines are of the same value (i.e., $k_1 = k_2 = \cdots = k_6 = k$), the stiffness matrix reduces to the form

$$K = kJ^{\mathrm{T}} J. \tag{6.74}$$

6.6.1 Stiffness Analysis of a 3–3 Stewart–Gough Platform

Figure 6.8 shows a schematic diagram of the 3–3 Stewart–Gough platform studied by Kerr (1989). There are six extensible limbs connecting a moving platform to a fixed base by two sets of concentric spherical joints located at points A_i and B_i, respectively. We wish to analyze the stiffness of this manipulator.

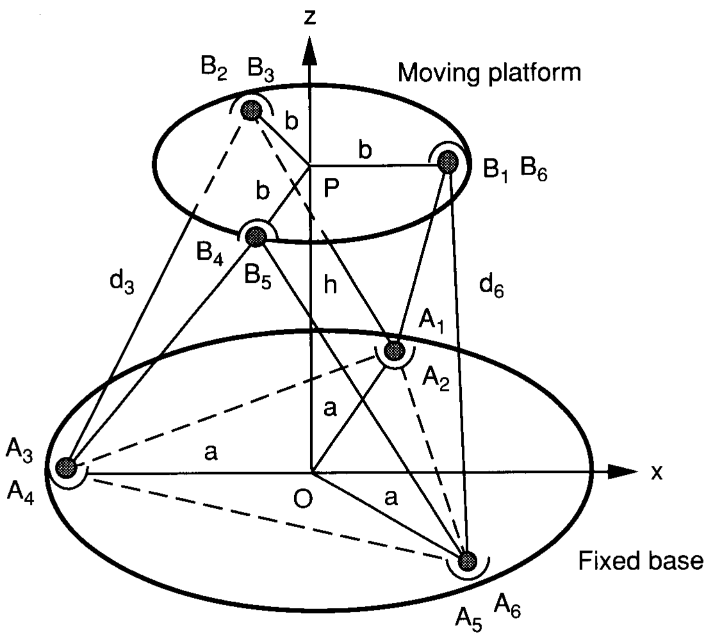

FIGURE 6.8. A 3–3 Stewart–Gough platform.

For the purpose of analysis, we attach an (x, y, z) coordinate frame to the fixed base with its origin O located at the centroid of the fixed base, its x and y axes lying on the plane of the attachment points, A_i, $i = 1, \ldots, 6$, its x-axis pointing in the direction opposite to $\overline{OA_3}$, and its z-axis pointing out of the plane of the attachment points, as shown in Fig. 6.8. Furthermore, we assume that the two sets of concentric spherical joints form two equilateral triangles. Since the stiffness matrix depends on the position and orientation of the moving platform, in what follows we assume that the moving platform is located at a *central configuration*, where the moving platform is not rotated with respect to the fixed base and the centroid of the moving platform is at an elevation $\overline{OP} = h$ above the centroid of the fixed base, as shown in Figs. 6.8 and 6.9.

From the geometry of the platform shown in Figs. 6.8 and 6.9, we obtain the vectors $\mathbf{a}_i$, $\mathbf{b}_i$, and $\mathbf{d}_i$ expressed in the fixed (x, y, z) frame as follows:

$$\mathbf{a}_1 = \mathbf{a}_2 = \left[\frac{a}{2}, \frac{\sqrt{3}\,a}{2}, 0\right]^{\mathrm{T}},$$

$$\mathbf{a}_3 = \mathbf{a}_4 = [-a, 0, 0]^{\mathrm{T}},$$

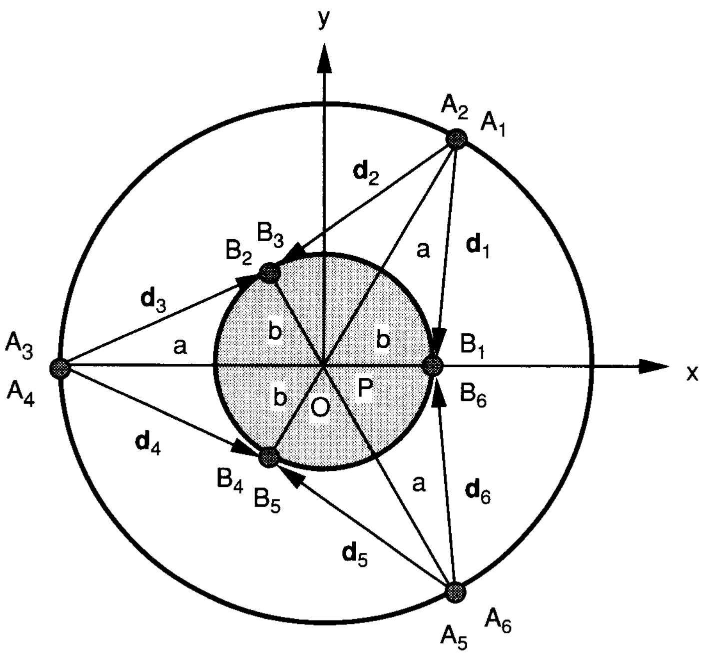

FIGURE 6.9. Top view of the 3–3 Stewart–Gough platform.

$$\mathbf{a}_5 = \mathbf{a}_6 = \left[\frac{a}{2}, -\frac{\sqrt{3}\,a}{2}, 0\right]^{\mathrm{T}},$$

$$\mathbf{b}_1 = \mathbf{b}_6 = [b, 0, 0]^{\mathrm{T}},$$

$$\mathbf{b}_2 = \mathbf{b}_3 = \left[-\frac{b}{2}, \frac{\sqrt{3}\,b}{2}, 0\right]^{\mathrm{T}},$$

$$\mathbf{b}_4 = \mathbf{b}_5 = \left[-\frac{b}{2}, -\frac{\sqrt{3}\,b}{2}, 0\right]^{\mathrm{T}},$$

$$\mathbf{d}_1 = \left[\frac{2b-a}{2}, -\frac{\sqrt{3}\,a}{2}, h\right]^{\mathrm{T}},$$

$$\mathbf{d}_2 = \left[\frac{-b-a}{2}, \frac{\sqrt{3}\,(b-a)}{2}, h\right]^{\mathrm{T}},$$

$$\mathbf{d}_3 = \left[\frac{-b+2a}{2}, \frac{\sqrt{3}\,b}{2}, h\right]^{\mathrm{T}},$$

$$\mathbf{d}_4 = \left[\frac{-b+2a}{2}, -\frac{\sqrt{3}\,b}{2}, h\right]^{\mathrm{T}},$$

$$\mathbf{d}_5 = \left[\frac{-b-a}{2}, \frac{\sqrt{3}\,(-b+a)}{2}, h\right]^{\mathrm{T}},$$

$$\mathbf{d}_6 = \left[\frac{2b-a}{2}, \frac{\sqrt{3}\,a}{2}, h\right]^{\mathrm{T}}.$$

At the central configuration, all the limbs are extended at equal lengths (i.e., $d_1 = d_2 = \cdots = d_6 = d$). This leads to a simple relation between a, b, h, and d:

$$d^2 = a^2 - ab + b^2 + h^2. \tag{6.75}$$

Substituting the expressions above into Eq. (6.68), we obtain

$$J = \frac{1}{2d} \begin{bmatrix} 2b-a & -\sqrt{3}\,a & 2h & 0 & -2bh & -\sqrt{3}\,ab \\ -b-a & \sqrt{3}\,(b-a) & 2h & \sqrt{3}\,bh & bh & \sqrt{3}\,ab \\ -b+2a & \sqrt{3}\,b & 2h & \sqrt{3}\,bh & bh & -\sqrt{3}\,ab \\ -b+2a & -\sqrt{3}\,b & 2h & -\sqrt{3}\,bh & bh & \sqrt{3}\,ab \\ -b-a & \sqrt{3}\,(-b+a) & 2h & -\sqrt{3}\,bh & bh & -\sqrt{3}\,ab \\ 2b-a & \sqrt{3}\,a & 2h & 0 & -2bh & \sqrt{3}\,ab \end{bmatrix}. \tag{6.76}$$

Substituting Eq. (6.76) and its transpose into (6.74), we obtain the stiffness matrix as

$$K = \frac{3k}{d^2} \times \begin{bmatrix} a^2+b^2-ab & 0 & 0 & 0 & bh(a/2-b) & 0 \\ 0 & a^2+b^2-ab & 0 & -bh(a/2-b) & 0 & 0 \\ 0 & 0 & 2h^2 & 0 & 0 & 0 \\ 0 & -bh(a/2-b) & 0 & b^2h^2 & 0 & 0 \\ bh(a/2-b) & 0 & 0 & 0 & b^2h^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3a^2b^2/2 \end{bmatrix}. \tag{6.77}$$

The upper left 3×3 submatrix represents the translational stiffness, the lower right 3×3 submatrix represents the torsional stiffness, and the other submatrices represent the cross-coupling effects between forces and moments, and between rotations and translations, respectively.

In practice, it is desirable to eliminate the cross-coupling terms. Fortunately, this can easily be achieved by choosing $a = 2b$ as a design condition. Substituting $a = 2b$ into (6.77) yields

$$K = \frac{3k}{d^2} \begin{bmatrix} 3b^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3b^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2h^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & b^2h^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & b^2h^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6b^4 \end{bmatrix}. \tag{6.78}$$

Observation of Eq. (6.78) reveals that three special designs are possible:

Case 1. $h^2 = 1.5b^2$ results in an isotropic translational stiffness, or
Case 2. $h^2 = 6b^2$ results in an isotropic torsional stiffness, or
Case 3. A compromise design between cases 1 and 2.

We note that without introducing additional design parameters, it is not possible to achieve both isotropic translational and isotropic torsional stiffness characteristics. Additional design parameters can be obtained by separating those concentric spherical joints located at either the moving platform or the fixed base.

6.7 SUMMARY

We have presented two methods of analysis, a free-body diagram method and the principle of virtual work, for the static analysis of serial and parallel manipulators. The free-body diagram method yields all the reaction forces acting on both the passive and actuated joints. While the principle of virtual work is more straightforward, it produces only the actuated joint forces. It was shown that in the absence of gravity, the actuated joint forces are related to the end-effector output forces by the transpose of the Jacobian matrix. The transformation characteristics were examined in detail by applying the theory of linear transformation. Furthermore, the stiffness characteristics of serial and parallel manipulators were examined. We have shown that the end-effector output forces are related to its deflections by a stiffness matrix. The stiffness matrix is a function of the stiffness constants of the drive lines and the Jacobian matrix of the manipulator. Since the Jacobian matrix is configuration dependent, the stiffness matrix is also configuration dependent. The statics and stiffness analyses of several manipulators were analyzed to illustrate the principles.

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EXERCISES

1. Figure 6.10 shows the planar five-bar $5R$ manipulator studied in Chapter 1. Calculate the actuator torques, τ_1 and τ_2, required to produce an output force of $(0, f_y)$ N at point Q. Investigate the conditions that will result in infinite input torques.

2. Figure 6.11 shows the pantograph mechanism studied in Chapter 1. Calculate the actuator forces, f_1 and f_2, required to produce an output force of (f_x, f_y) N at point Q.

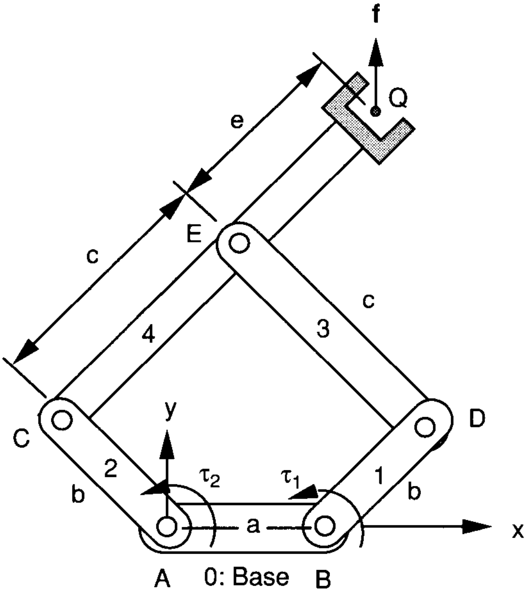

FIGURE 6.10. Statics of a five-bar manipulator.

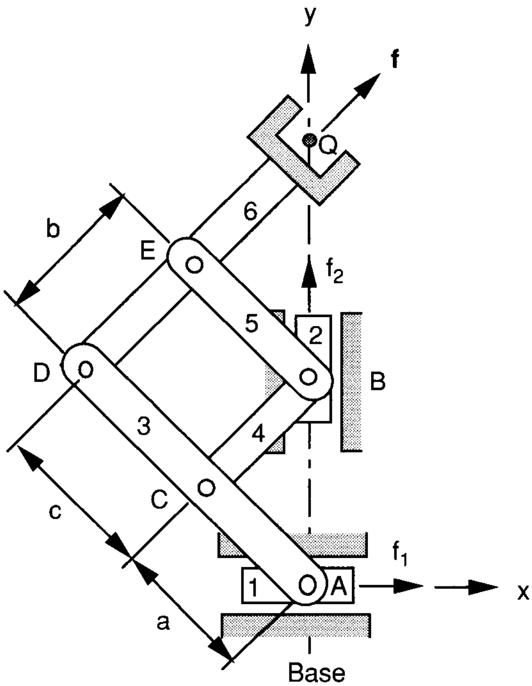

FIGURE 6.11. Statics of a pantograph.

3. Using the free-body diagram method, determine the actuator torques and/or forces required to produce an end-effector output force of $(0, 0, f_z)$ N at point Q of the SCARA arm shown in Fig. 2.4. Express these torques as functions of joint angles.

4. Using the free-body diagram method, find the actuator forces required to produce an output force of $(f_x, 0)$ N at the centroid of the planar $3RPR$ parallel manipulator shown in Fig. 6.12. Let the prismatic joints be the actuated joints.

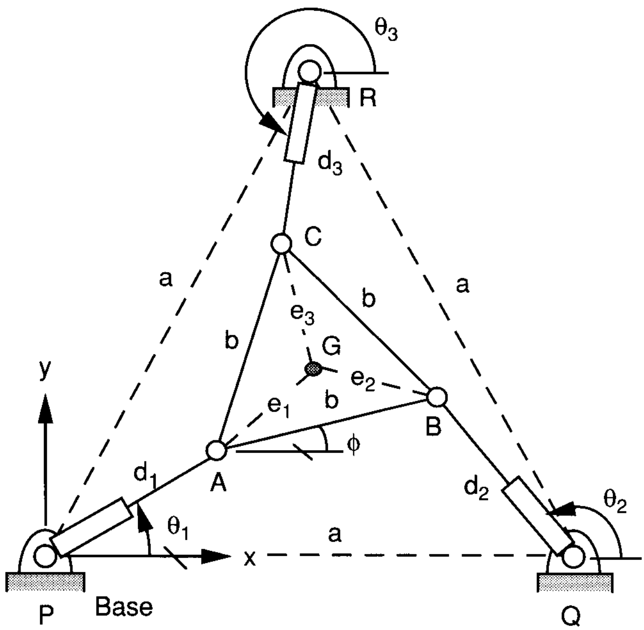

FIGURE 6.12. Schematic diagram of a planar $3RPR$ manipulator.

5. For the DELTA robot shown in Fig. 3.21, let the moving platform be located directly on top of the fixed platform. Also let the sizes of the two platforms be the same. Calculate the joint torques required to produce an end effector output force of $(0, 0, f_z)$ N. Show that the actuator torques tend to infinity as the moving platform approaches the fully stretched location.

6. Calculate the actuator torques required to generate an end-effector output force $\mathbf{f} = (f_x, f_y)$ N at point Q of the planar five-bar $4R1P$ manipulator shown in Fig. 3.19 by the principle of virtual work.

7. Calculate the joint torques required to produce an end-effector output force $\mathbf{f} = (f_x, f_y, f_z)$ N at point Q of the SCARA arm shown in Fig. 2.4 by the principle of virtual work.

8. Derive the stiffness matrix of the planar five-bar $5R$ manipulator shown in Fig. 6.10. Under what conditions will the manipulator lose its stiffness?

9. Derive the stiffness matrix of the manipulator shown in Fig. 5.15, assuming that the moving platform is located at a central location. Under what conditions will the stiffness matrix become decoupled?